A Machine-Checked Audit of the Extended-Extraction RG Contraction in the Wilson-Lattice / OS Route to a 4D Yang–Mills Mass Gap

The MeTTapedia formalization effort

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Abstract

Ben Goertzel's manuscript *Yang–Mills Mass Gap in 4D: RG Bootstrap, Polymer Expansion, OS Reconstruction* (v14) proposes to construct a continuum SU(N) Yang–Mills theory from Wilson lattice gauge theory via a multiscale renormalization-group (RG) bootstrap, a Kotecký–Preiss (KP) polymer expansion, exponential clustering, and Osterwalder–Schrader (OS) reconstruction. Its technical keystone is an *extended-extraction* contraction $\Lambda = C_1\lambda_{\mathrm{irr}} < 1$, with advertised constant near 2224 and $\lambda_{\mathrm{irr}} = 2^{-13}$ at block factor 2 and extraction depth 16.

The scalar bootstrap arithmetic is exact, but the advertised Wilson recursion constant is *not derived in v14 as written*. The main-text factor product is 33152 at $b = 2$; Appendix O's different proposed factorization is 11088/5 when its extraction factor is 2. A sign-only repair of the printed extraction series is insufficient once the displayed multivariate multiplicities are retained.

We construct a concrete finite Wilson-polynomial realization on actual group-valued link fields. In its explicit coefficient-majorant norm, canonical-dimension truncation is gauge invariant and has operator constant 1. This does not transfer from v14's range-only specification of P_{ext}: a skew projection onto the same low-coordinate range has arbitrarily large coefficient norm. We prove the correct transfer theorem from explicit basis and dual-jet matrices with their column-sum conditioning bounds, but v14 does not provide those data for the dimension-16 action sector. A direct source reconciliation shows that they cannot be reconstructed uniquely from the present text: Appendix F.4 uses equations of motion to identify the operator space, while Appendix O.9 explicitly retains EOM operators in an off-shell basis; neither section supplies lattice Wilson representatives or a quotient basis, and Appendix I's explicit dual jets stop at the distinct dimension-4 observable sector. Likewise, regrouping connected families by union support has norm 1; the factor 14 is exactly a *local rooted branching count* $2(2 \cdot 4 - 1)$, not a bound on the total contacts of two extended polymers. The fluctuation big-O and weighted cumulant-generation estimates also require quantitative proofs on the actual Wilson RG map.

A third machine-checked round works on the exact SU(2), $b = 2$ Wilson block with normalized Haar measure and decides the extraction stage from three sides. (i) Representative-wise hard truncation is not well-defined on the actual trace-word quotient: for the centered trace words $a = 2 - \mathrm{Tr}(U)$ and $b = 2 - \mathrm{Tr}(U^2)$, Cayley–Hamilton forces $a^3b - 4a^4 + a^5 = 0$ with monomial canonical dimensions 16, 16, 20; truncation at 16 deletes only a^5, and the truncated expression evaluates to -32 on the holonomy $\mathrm{diag}(i, -i)$ while the exact expression vanishes identically. (ii) Two corrected constructions nevertheless carry exact norm-below-two extractors — the full associated-graded jet chart ($131071/65536 < 2$) and the one-plaquette trace quotient ($31/16 < 2$); with Appendix O's other factors kept unchanged, the resulting conditional constant $90832203/40960 \approx 2217.6$ closes the two-source bootstrap at depth 16 and fails at depth 15. (iii) Neither survivor transfers to v14's canonical (F, D) operator space: a dimension-18 scalar descendant has radial degree only 6 (resp. 3), so both extractors retain it, and the dimension-6

descendant $\partial^2 \mathrm{Tr}(F^2)$ is zero in the manuscript's integration-by-parts quotient yet has a nonzero gauge-invariant realization on an explicit three-plaquette block. The transfer socket required before any Wilson-RG intertwiner is therefore uninhabited as written, under both the on-shell and off-shell relation policies.

A fourth machine-checked round tests the replacement census against the exact symmetry of the finite-spacing lattice. On an explicit SU(2) Cartan link chart, the sum of squared normalized Wilson potentials over the six coordinate plaquettes is gauge invariant and has canonical dimension 8. Its exact fourth jet is $6 \sum_{\mu<\nu} F_{\mu\nu}^4$. A rational orthogonal rotation changes the corresponding quartic from 1 to 337/625. Thus the observable is hypercubic-invariant but not $O(4)$-invariant. Every span of Appendix O.9.3's Lorentz-scalar contractions remains $O(4)$-invariant, so it cannot be complete for actual lattice Wilson observables. This obstruction closes the off-shell, on-shell EOM, Bianchi/IBP, trace, Cayley–Hamilton, and postponed-IBP variants of the same Lorentz-only target. It does not rule out a finite coordinate system: it forces either a genuinely hypercubic census at finite lattice spacing or a separately proved continuum/Symanzik projection with a quantitatively controlled Lorentz-breaking remainder.

A fifth machine-checked round begins the hypercubic branch. On the unpadded derivative-free Cartan quartic carrier, all 126 monomials are classified by signed $H(4)$ orbit certificates. Exactly four admissible orbit basis functions remain; evaluation at their representatives gives four duals and the identity conditioning matrix, and every hypercubic-covariant rational quartic coefficient function is reconstructed from them. The exact Wilson fourth jet has coordinate vector $(6, 0, 0, 0)$ in this basis. The subspace of curvature observables invariant under every $O(4)$ transformation is proper, and the pure-fourth Wilson polynomial is a nonzero class in the corresponding Lorentz-breaking quotient. The attempted dimension-16 extension then reaches a different terminal obstruction: its fixed-width raw carrier retains inactive padding as part of equality. Two distinct encodings with zero active fields and derivatives have identical active syntax. Any basis faithful to active syntax must give them equal rows, whereas the proposed exact census requires a right inverse and hence distinct identity-matrix rows. Lean proves that no invertible exact census on this padded carrier can have every basis column padding-invariant. The next finite-spacing construction must use an unpadded dependent syntax or an explicit padding quotient before relation rows and dimension-16 duals are meaningful.

Thus there is presently no derived constant C_{true} for the Wilson recursion. The support recurrence, two-marked glue identity, stopping-scale polymer estimates, Wilson reflection positivity, and continuum/OS construction remain route-specific open inputs. The finite $\mathbb{Z}_2$, $\mathbb{Z}_3$, and Q_8 gaps checked in Lean are real but Clay-irrelevant (single-link, strong-coupling, finite). The Yang–Mills mass gap remains open; we make no claim about it.

1 The problem and Ben Goertzel's route

1.1 The Millennium problem and the lattice strategy

Fix the gauge group $G = \mathrm{SU}(N)$ with $N \geq 2$ and let $\mathbb{R}^4$ be Euclidean space. The Yang–Mills mass-gap problem asks for the construction of a nontrivial continuum quantum Yang–Mills theory on $\mathbb{R}^4$ with gauge group G, satisfying the Wightman / Osterwalder–Schrader axioms, whose reconstructed Hamiltonian has a unique vacuum and a strictly positive spectral gap on the orthocomplement of the vacuum.

Goertzel's v14 manuscript follows Wilson's lattice regularization. One discretizes $\mathbb{R}^4$ as a hypercubic lattice $a\mathbb{Z}^4$ with spacing $a > 0$, with link variables $U_{x,\mu} \in G$ on oriented nearest-neighbor links. The plaquette variable is the parallel transporter around an elementary square,

$$U_{x,\mu\nu} \;=\; U_{x,\mu}\, U_{x+ae_\mu,\nu}\, U_{x+ae_\nu,\mu}^{-1}\, U_{x,\nu}^{-1}, \qquad 1 \leq \mu < \nu \leq 4,$$

and the Wilson action penalizes deviation of plaquettes from the identity,

$$S_{\Lambda,a}(U) \;=\; \beta \sum_{p\subset\Lambda} V(U_p), \qquad \beta = \frac{2N}{g^2}, \qquad V(U) := 1 - \tfrac{1}{N}\mathrm{Re}\,\mathrm{Tr}(U) \in [0,2].$$

Weak coupling is $\beta \gg 1$. The strategy is: (Step 1) start from the well-defined finite-dimensional lattice integral; (Step 2) run a multiscale RG bootstrap controlling the effective theory across all scales; (Step 3) establish exponential clustering at a fixed physical scale ℓ_* via a KP polymer expansion; (Step 4) transfer clustering from the coarse scale back to the fine scale with a glue / back-propagation lemma; (Step 5) reconstruct the continuum Hilbert space, vacuum, and Hamiltonian by OS, with the clustering rate becoming the spectral gap; (Step 6) verify nontriviality through a nonzero third cumulant of the action density.

1.2 The continuum-limit obstruction and extended extraction

In any RG scheme one must show that the "irrelevant" sector (operators of canonical dimension > 4) contracts at each step:

$$\|K^{\mathrm{irr}}_{j+1}\| \;\le\; \Lambda\,\|K^{\mathrm{irr}}_{j}\| + (\text{source terms}), \qquad \Lambda < 1.$$

The manuscript identifies a sharp obstruction in the *naive* scheme, where one extracts only the relevant and marginal operators of dimension ≤ 4. Under rescaling, a dimension-d operator scales by b^{4-d}, so the leading irrelevant operators (dimension 6) contract by $b^{-2} = 0.25$ for $b = 2$. But the RG step also recombines polymers, contributing a combinatorial constant C_1. The manuscript advertises the working bound $C_1 \le 2224$ for $b = 2$; if that value were established, the naive gain would be

$$\Lambda_{\mathrm{naive}} \;=\; 2224\,b^{-2} \;=\; 2224 \times 0.25 \;=\; 556 \;>\; 1,$$

an *expanding* map: initial smallness would grow like 556^J over $J \sim |\log a_0|$ steps and diverge as $a_0 \to 0$.

The proposed resolution is *extended extraction.* Instead of extracting only dimension ≤ 4 operators, one extracts all gauge-invariant local operators up to canonical dimension $d_{\max} = 16$ (for $b = 2$). The irrelevant sector then begins at dimension $d_{\max} + 1 = 17$ and contracts by

$$\lambda_{\mathrm{irr}} \;=\; b^{\,4-(d_{\max}+1)} \;=\; b^{\,3-d_{\max}} \;=\; 2^{-13} \;\approx\; 1.2 \times 10^{-4}.$$

With the same advertised value, the proposed contraction factor becomes

$$\boxed{\Lambda \;=\; C_1\,\lambda_{\mathrm{irr}} \;\le\; 2224 \cdot 2^{-13} \;\approx\; 0.27 \;<\; 1.}$$

The manuscript presents this as genuine contraction "with substantial slack." The audit below verifies this arithmetic only *conditional on the supplied constant*; it refutes the available derivations of that constant as written. The price is tracking a finite-dimensional vector of higher couplings $\vec{g}_j = (g_j^{(4)}, g_j^{(6)}, \ldots, g_j^{(d_{\max})})$ rather than the single gauge coupling; the manuscript argues these are perturbatively small and do not affect the universal one-loop beta-function coefficient $b_0 = 11/(3 \cdot 16\pi^2)$, which is determined entirely by local ultraviolet information.

1.3 The recombination constant and the extraction-projection norm

Appendix O proposes the factorization

$$C_1 \;\le\; C_{\text{fluct}} \cdot C_{\text{cumulant}} \cdot C_{\text{recomb}} \cdot C_{\text{extract}} \;\le\; 1.65 \times 3b^4 \times 14 \times 2,$$

which at $b = 2$ equals $11088/5 = 2217.6$, rather than exactly 2224. Appendix F.5 displays a different factor product which equals 33152 at $b = 2$; the two ledgers are inconsistent. The last Appendix-O factor $C_{\text{extract}} = \|P_{\le d_{\max}}\|_{\text{op}}$ is the operator norm of the extraction projection onto the finite-dimensional space of operators of dimension $\le d_{\max}$. The manuscript bounds it (its Computation O.7(d)) by

$$\|P_{\le d_{\max}}\|_{\text{op}} \;\le\; \sum_{d=0}^{d_{\max}} \binom{\#\text{indices}}{d} (r_1/r_0)^{-d} \;\le\; 1 + O((r_0/r_1)^{d_{\max}}),$$

and asserts $\|P_{\le d_{\max}}\|_{\text{op}} \le 2$ for $r_1/r_0 = 1/2$ and $d_{\max} = 16$. The literal negative exponent grows, but changing only its sign does not turn the displayed multivariate sum into a one-dimensional geometric series. More fundamentally, v14 specifies P_{ext} by its range and does not give the dimension-16 invariant basis, dual jets, or conditioning estimates needed to determine its operator norm. We return to the proved realization and the transfer obstruction in §3.

2 What we formalized

The formalization lives in the Lean namespace `Mettapedia.QuantumTheory.YangMills`. We summarize the four checked components: the contraction predicate, the knife-edge arithmetic, the finite strong-coupling gaps with their finite OS reconstruction, and the continuum-open gating. All theorem and definition names below are reproduced verbatim from the corresponding modules.

2.1 The extended-extraction contraction predicate

The module `Mettapedia.QuantumTheory.YangMills.RGBootstrap` models the route's contraction obligation as a predicate over a recombination constant C, block factor b, and extraction depth $d_{\max}$. The irrelevant scaling factor is $b^{3-d_{\max}}$ (an integer power, since the useful cases have $d_{\max} > 3$), and a contraction is a nonnegative constant whose product with the irrelevant scale is below 1.

```
/-- Irrelevant scaling factor ‘b^(3-dmax)‘ used by the extended-extraction
discussion.  The exponent is an integer because the useful cases have
‘dmax > 3‘. -/
def irrelevantScale (b dmax : ) :  :=
  (b : ) ^ (3 - (dmax : ))

/-- The route-level contraction audit for a combinatorial constant and an
irrelevant scaling factor. -/
def HasRGContraction (C scale : ) : Prop :=
  0  C  0  scale  C * scale < 1

/-- The extended-extraction audit specialized to ‘ = b^(3-dmax)‘. -/
def HasExtendedExtractionContraction (C : ) (b dmax : ) : Prop :=
  HasRGContraction C (irrelevantScale b dmax)
```

A threshold lemma in the same module records the exact content of the predicate: for $b > 0$ and $d_{\max} \geq 3$, the contraction `HasExtendedExtractionContraction C b dmax` holds if and only if $0 \leq C < b^{d_{\max}-3}$. Thus the entire arithmetic content of the route is the comparison of C with the threshold $b^{d_{\max}-3}$.

2.2 The knife-edge arithmetic

The module machine-checks that the advertised constant contracts at $d_{\max} = 16$ and that the exact one-step gain is $2224 \cdot 2^{-13} = 139/512 < 1/3$:

```
/-- The advertised numerical slack: ‘2224 * 2^-13 < 1‘. -/
theorem rgContraction_2224_two_sixteen :
    HasExtendedExtractionContraction 2224 2 16 := by
  refine by norm_num, by rw [irrelevantScale_two_sixteen]; norm_num, ?_
  rw [irrelevantScale_two_sixteen]
  norm_num

/-- Exact advertised one-step gain for the extended-extraction constants. -/
theorem rgGain_2224_two_sixteen_eq :
    (2224 : ) * irrelevantScale 2 16 = (139 : ) / 512 := by
  rw [irrelevantScale_two_sixteen]
  norm_num
```

Crucially, the same constant does *not* contract at the nearest even extraction depth below the advertised one. At $d_{\max} = 14$ the threshold is $2^{11} = 2048 < 2224$, so the contraction is arithmetically false; the module proves that any contraction at $d_{\max} = 14$ would force $C < 2048$, and hence that 2224 fails:

```
/-- Any contraction proof at ‘dmax = 14‘ must improve the linear recombination
constant below ‘2048‘.  The manuscript’s advertised bound ‘2224‘ is therefore
insufficient for this lower extraction depth. -/
theorem rgContraction_two_fourteen_forces_constant_lt_2048
    {C : } (h : HasExtendedExtractionContraction C 2 14) :
    C < 2048 := by
  have hprod : C * irrelevantScale 2 14 < 1 := h.2.2
  rw [irrelevantScale_two_fourteen] at hprod
  nlinarith

/-- Same-constant blocker at the nearest even extraction depth below ‘16‘. -/
theorem not_rgContraction_2224_two_fourteen :
    ¬ HasExtendedExtractionContraction 2224 2 14 := by
  intro h
  have hC : (2224 : ) < 2048 :=
    rgContraction_two_fourteen_forces_constant_lt_2048 h
  norm_num at hC
```

This is the knife edge: the choice $d_{\max} = 16$ is exactly what makes the advertised constant contract, and the manuscript's own table shows $d_{\max} \in \{10, 12, 14\}$ all fail against $C_1 \leq 2224$ while $d_{\max} = 16$ succeeds. The module `Mettapedia.QuantumTheory.YangMills.RGCrux` packages this into a route-level statement: a mass-gap route that keeps the advertised constant 2224 but uses an even extraction depth strictly below 16 cannot exist, since the required contraction is arithmetically false before any OS / clustering layer is reached.

```
/-- Same-constant lower-extraction certificate in the manuscript's even
canonical-dimension shape: the cutoff is strictly below ‘16‘, is even, and uses
the same advertised recombination bound ‘2224‘. -/
def SameConstantEvenBelowSixteenRGCertificate (dmax : ) : Prop :=
  dmax < 16  Even dmax  HasExtendedExtractionContraction 2224 2 dmax

/-- No same-constant RG certificate exists for an even manuscript-shape cutoff
strictly below ‘16‘. -/
theorem not_sameConstantEvenBelowSixteenRGCertificate
    {dmax : } :
    ¬ SameConstantEvenBelowSixteenRGCertificate dmax := by
  intro hcert
  have hdmax : dmax  14 :=
    even_lt_sixteen_le_fourteen hcert.1 hcert.2.1
  exact (not_rgContraction_2224_two_of_dmax_le_fourteen hdmax) hcert.2.2
```

A strengthened companion theorem, `not_atLeast2048EvenBelowSixteenRGCertificate`, shows that any even lower-extraction repair below $d_{\max} = 16$ would need a genuinely sharper constant ($C < 2048$), not merely the advertised one. We stress that this is an honest delimiter of the arithmetic surface, not a refutation of the route: at the raw natural number $d_{\max} = 15$ the same constant *does* contract ($2^{12} = 4096 > 2224$), and the module records this explicitly so that the obstruction is read precisely as an even canonical-dimension statement matching the manuscript's dimensions $4, 6, \ldots, d_{\max}$.

2.3 Finite strong-coupling spectral gaps and finite OS reconstruction

The module `Mettapedia.QuantumTheory.YangMills.FiniteOSReconstruction` formalizes the finite algebraic core of OS reconstruction: a reflection-positive finite kernel induces a positive pairing on observables, and a transfer operator that is self-adjoint for that pairing descends to the pairing quotient and remains self-adjoint there. On this base, the modules `Z2StrongCouplingGap` and `Z3StrongCouplingGap` build explicit finite single-link abelian lattice gauge models. For $\mathbb{Z}_2$, the one-link heat-kernel transfer at heat time 1 has eigenvalues 1 and $\exp(-1)$, so the dimensionless gap is $-\log(\exp(-1)/1) = 1$:

```
theorem z2StrongCouplingGap_eq_one :
    z2StrongCouplingGap = 1 := by
  simp [z2StrongCouplingGap, z2StrongCouplingLambdaTwo,
    z2StrongCouplingLambdaOne, z2StrongCouplingRatio, Real.log_exp]

/-- End-to-end finite lattice theorem: the explicit ‘Z‘ strong-coupling model is
nontrivial, reflection positive, has an OS transfer operator, and has
dimensionless transfer gap at least ‘1‘. -/
theorem z2StrongCoupling_latticeYangMills_massGap_landmark :
     _ : Z2StrongCouplingGapCanary,
      ( x y : Z2OneLinkConfig, x  y)
        OSReflectionPositive z2StrongCouplingKernel
        z2StrongCouplingGap = 1
        (1 : )  z2StrongCouplingGap
        0 < z2StrongCouplingGap := by
  exact
    z2StrongCouplingPositiveCanary,
```

```
      z2OneLinkConfig_nontrivial,
      z2StrongCouplingKernel_reflectionPositive,
      z2StrongCouplingGap_eq_one,
      z2StrongCouplingGap_lower_bound_one,
      z2StrongCouplingGap_positive
```

The $\mathbb{Z}_3$ module proves the analogous landmark with two mean-zero transfer eigenvectors and the same unit gap. These theorems are unconditional and non-vacuous: they exhibit genuine reflection positivity, a genuine finite OS transfer reconstruction, and a strictly positive spectral gap. They are also kept rigorously separate from any continuum claim. Each module carries an explicit scaling diagnostic showing that a finite gap need not survive a naive continuum schedule. For the heat-time schedule $t(a) = a^2$, the *physical* gap (dimensionless gap divided by lattice spacing) closes as $a \to 0$:

```
/-- A concrete gap-closing schedule for the finite model under the heat-time
scaling ‘t(a)=a^2‘: the physical gap can be made smaller than any positive
threshold by taking the lattice spacing small enough. -/
theorem z2QuadraticHeatTime_physicalGap_closes :
      : , 0 <
       a : ,
        0 < a  a <
          z2HeatTimePhysicalGap a (a ^ 2) <  := by
  intro  h
  refine  / 2, by linarith, by linarith, ?_
  have hne :  / 2  0 := by linarith
  rw [z2QuadraticHeatTime_physicalGap_eq hne]
  linarith
```

2.4 The continuum endpoint is gated open

The module `Mettapedia.QuantumTheory.YangMills.ProofState` records a status ledger over the route's nodes. The finite RG / extraction surfaces are marked `checked` or `refuted` as appropriate, but the continuum mass-gap endpoint carries an explicit `openGoal` status, gated behind a constructive-QFT promotion barrier (continuum measure, Euclidean covariance, reflection positivity, OS reconstruction, and Hamiltonian mass-gap transfer — none of which is represented for the RG / extraction route, which supplies only a finite ledger):

```
/-- The final mass-gap endpoint remains an open goal in this lane. -/
def yangMillsMassGapEndpointNode : YangMillsProofNode where
  id := "yang-mills.mass-gap-endpoint"
  status := .openGoal
  truthValue := 0, 95
  evidence := "currentYangMillsMassGapEndpoint_blockedByConstructiveGate blocks promotion
     from the audited RG/extraction surfaces to a continuum mass-gap endpoint ..."
  nextObligation := "Connect the RG and extraction surfaces to a genuine continuum mass-
    gap statement only after the constructive-QFT gate, OS/Hamiltonian transfer
    dependency packet, challenge-world interface packet, and mass-gap promotion gate
    clear."

theorem yangMillsMassGapEndpointNode_open :
    yangMillsMassGapEndpointNode.status = .openGoal := by
  rfl
```

A companion theorem `currentYangMillsMassGapEndpoint_blockedByConstructiveGate` records that the endpoint is open, the constructive checks are all unrepresented, and the only supplied interface is the finite RG / extraction ledger. In short, the formalization does not overclaim: nothing in it asserts a continuum mass gap.

3 Honest status and scope

This lane is, by construction, conservative; we now state precisely what is and is not established.

The contraction constant is an input, not a conclusion. Every arithmetic theorem above takes the recombination constant as a *given real number*. The predicate `HasExtendedExtractionContraction C b dmax` is a statement about C, b, $d_{\max}$ alone; the checked facts `rgContraction_2224_two_sixteen` and `not_rgContraction_2224_two_fourteen` verify the comparison $C \leqslant b^{d_{\max}-3}$ for the specific value $C = 2224$. Nothing in the formalization derives that 2224 (or any $O(1) \cdot b^4$ bound) is the correct value of the manuscript's recombination constant. The Lean ledger is therefore faithful to the route's bookkeeping but agnostic about the route's hardest analytic claim.

The extraction projection: realization succeeds, range-only transfer fails. The sign-only repair is insufficient: even with positive exponent, the displayed binomial/multivariate multiplicities make the corrected finite sum exceed 2 once at least two coordinates are present. We therefore constructed the natural finite polynomial realization rather than treating the omission as a verdict. For a finite Wilson block of actual group-valued links, plaquette holonomies, and a conjugation-invariant Wilson potential, finite polynomials in the plaquette coordinates are gauge invariant. In the explicitly defined weighted coefficient-majorant norm, canonical-dimension truncation has operator constant 1 (module `WilsonBlockMajorant`).

This is our realization of the manuscript's extraction idea, not an attribution to v14. It cannot be transferred to v14's P_{ext} from the specified range alone: `projectionRangeSpecification_does_not_imply_bound_two` constructs an idempotent skew projection onto the same low-coordinate range that violates the bound 2. The positive transfer theorem `coefficientL1_basisJetProjection_le` shows exactly what would suffice: explicit basis and dual-jet matrices whose coefficient-column bounds multiply. The construction attempt is certified in `Dimension16WilsonExtractionCertificate`: F.4's on-shell relation policy and O.9's off-shell policy disagree already on the displayed dimension-6 divergence-square operator; the exact block quotient, lattice representatives, finite basis, dimension-16 dual jets, biorthogonality, and conditioning matrices are not supplied. Appendix I's explicit jets concern the separate observable projection $P^{\text{obs}}_{\leq 4}$. Hence the extraction component is not a standing erratum repair, and C_1 is not derived.

Hard truncation does not descend to the actual trace-word quotient. The third round replaces schematic carriers by the exact object: the SU(2), $b = 2$ Wilson block with normalized Haar measure — 81 vertices, 216 positive links of which 8 are internal, 216 plaquettes, an internal spanning tree with 7 co-tree links, exact boundary-plus-co-tree coordinates, the gauge action, the normalized Wilson potential, and the fixed-boundary product-Haar reference integral with its logarithmic effective block action (modules `SU2WilsonBlockPilot`, `SU2MajorantRepairDecision`). On this object the question "is canonical-dimension truncation even well-defined on the gauge-invariant quotient?" has a negative, intrinsic answer. For any $U \in \mathrm{SU}(2)$ the Cayley–Hamilton identity gives $\mathrm{Tr}(U^2) = \mathrm{Tr}(U)^2 - 2$; writing $a = 2 - \mathrm{Tr}(U)$ and $b = 2 - \mathrm{Tr}(U^2)$ (each of canonical dimension 4), the exact relation $b = 4a - a^2$ yields

$$a^3 b - 4a^4 + a^5 \;=\; 0,$$

whose three monomials have construction-derived canonical dimensions 16, 16, 20. Truncation through dimension 16 deletes only a^5; on the explicit holonomy $\mathrm{diag}(i, -i)$ the truncated expression evaluates to -32, while the exact left-hand side vanishes on every field. Hence the semantic kernel is not stable under representative-wise hard truncation, no function on the evaluated trace-word quotient agrees with that truncation, and the corresponding descended extraction certificate is uninhabited (module `SU2CayleyHamiltonTruncationObstruction`). The witness uses only the Cayley–Hamilton relation — no equations-of-motion quotient and no boundary-extension convention — so it applies to both of v14's relation policies, and it refutes any uniform-in-N architecture that requires representative-wise hard truncation of redundant Wilson trace words.

Corrected constructions do carry norm-below-two extractors. Truncation is, of course, definable on a *chosen basis*, and the third round constructs the two natural repairs exactly. The full associated-graded jet chart on the gauge-fixed block (7 co-tree links, 3 Lie coordinates each, hence 21 coordinates) carries the symmetric-monomial basis of jets through total degree 16, with identity jet matrix and biorthogonal coefficient duals; radial Taylor extraction at radius ratio 1/2 has the exact operator constant $\sum_{d=0}^{16} 2^{-d} = 131071/65536 < 2$ (modules `SU2AssociatedGradedExtraction`, `SU2AssociatedGradedExtractionDecision`). Independently, the evaluated one-plaquette trace algebra is exactly a one-variable polynomial algebra: the normal-form theorem $b = 4a - a^2$ (proved semantically complete via an infinite Cayley-rotation family) gives the basis $1, a, a^2, a^3, a^4$ through dimension 16, with extraction constant $\sum_{d=0}^{4} 2^{-d} = 31/16 < 2$ (module `SU2IndependentTraceBasis`). Keeping Appendix O's other factors unchanged, the associated-graded constant yields the conditional ledger value

$$C_{\text{proposed}} \;=\; \tfrac{90832203}{40960} \;\approx\; 2217.583 \;<\; 2224,$$

whose two-source gain at depth 16 is $90832203/335544320 \approx 0.2707 < 1/3$, while depth 15 does not close. This is conditional bookkeeping: the fluctuation, cumulant, and recombination factors remain unproved on the actual map. Two exact theorems keep these constructions honestly separated from v14's object: the full graded chart has dimension 12,875,774,670 and admits no linear equivalence onto any coordinate space of dimension at most 1851, so the manuscript's schematic count requires a genuine invariant quotient that v14 does not construct; and the radial norm used here is a precise repair, not provably the manuscript's intended multivariate norm.

Neither survivor transfers to the canonical (F, D) space. The transfer question is then decided negatively by two independent machine-checked obstructions (module `V14FDQuotientTransferNoGo`, headline `v14_fullFD_quotientTransfer_noGo`). *Derivative blindness:* v14 counts canonical dimension as $2 \cdot \#F + \#D$ (its O.9.1), but radial degree on either chart is blind to derivative factors. The scalar descendant $(\partial^2)^3[(\mathrm{Tr}F^2)^3]$ has canonical dimension 18 yet radial degree 6 on the associated-graded chart and 3 on the trace chart, so both norm-below-two extractors *fix* it — and an explicit two-cell zero-boundary sixth-difference witness evaluates to -6, so the retained operator is nonzero. Both survivors therefore fail to implement the dimension-$\leq$ 16 projection of F.4.2, independently of the choice of quotient basis (`not_associatedRadialImplementsCanonicalCutoff`, `not_independentRadialImplementsCanonicalCutoff`). *Integration-by-parts/block incompatibility:* v14's local norm (F.3) lives on ordinary functions on a finite block retaining boundary data, while both relation branches identify integrated total derivatives with zero. The dimension-6 descendant $\partial^2 \mathrm{Tr}(F^2)$ is gauge invariant and evaluates to 1 on an explicit three-plaquette block field (a purely quadratic associated-grade witness gives the same value, so higher Taylor terms are not the cause). An ordinary block-functional realization would have to send one quotient class to both the zero function and a nonzero boundary descendant; the faithful transfer sockets — exact-Wilson, pure associated-graded, and every RG-compatible extension — are therefore uninhabited under both relation policies. Consequently no corrected recursion constant propagates from the constructions above into the bootstrap. The obstruction is structural rather than informational: supplying the omitted dimension-16 coordinates or conditioning matrices cannot make either extractor into the canonical (F, D) projection. A viable replacement needs a derivative-aware filtration whose truncation provably agrees with $2 \cdot \#F + \#D$, together with a boundary-aware complex that retains total-derivative descendants on each block as boundary cochains and imposes integration by parts only at global recombination; constructing exactly that replacement is the current object of this formalization.

The actual-lattice census cannot be Lorentz-only. The derivative-aware boundary-cochain repair makes the filtration and local boundary bookkeeping explicit, but a further obstruction appears before its coordinate basis or dual jets can be supplied. Appendix O.9.3 enumerates only Lorentz-invariant complete contractions. The exact finite-spacing Wilson lattice has only hypercubic symmetry, and the module `SU2LatticeFDCensusNoGo` supplies a gauge-invariant canonical-dimension 8 observable whose fourth Cartan jet is

$$6(F_{01}^4 + F_{02}^4 + F_{03}^4 + F_{12}^4 + F_{13}^4 + F_{23}^4).$$

An explicit rational orthogonal rotation changes its normalized quartic value from 1 to 337/625. Hence no finite span of the $O(4)$-invariant O.9.3 jets contains it. The necessary spanning core, the O.9.2 off-shell certificate, and the F.4.3 on-shell certificate are all formally uninhabited. The witness is derivative-free, so changing EOM, Bianchi, integration-by-parts, trace, Cayley–Hamilton, or boundary-cochain reduction policies cannot repair this same-target symmetry mismatch. The result is already an $N = 2$ counterexample to an architecture asserted uniformly for $\mathrm{SU}(N)$. It is not a no-go for a hypercubic-complete basis or for a controlled continuum projection.

The first hypercubic census succeeds at quartic order and refutes its padded extension. The unpadded carrier of degree-four monomials in the six oriented two-form components has cardinality 126. Exact signed-orbit certificates divide it into four admissible $H(4)$ orbit types — a pure fourth power, adjacent squares, opposite squares, and a rectangle product — plus monomials killed by negative stabilizers. Four signed orbit indicators obey the full hypercubic covariance law, their representative evaluations form the 4×4 identity, and they reconstruct every invariant rational coefficient function. This is a complete result for the derivative-free Cartan quartic restriction, not for the noncommutative (F, D) quotient. The actual gauge-invariant Wilson observable has exact fourth derivative equal to the polynomial realization of coordinate vector $(6, 0, 0, 0)$. Moreover, the curvature observables invariant under every orthogonal spacetime matrix form a proper linear subspace, and this pure-fourth polynomial defines a nonzero class in the quotient by that subspace. Thus the Lorentz-breaking artifact is a coordinate-level theorem, not merely an external rotation counterexample.

The fixed-width full carrier cannot supply that extension faithfully. Inactive field and derivative slots are mathematically padding but remain part of raw equality. The module

`V14HypercubicFDCensusPaddingNoGo` constructs two distinct encodings of the empty active syntax and proves `faithful_dimension16_exactCensus_uninhabited`: if every basis column ignores inactive padding, the two basis rows coincide; the required right inverse instead makes them two different rows of the identity matrix. This obstruction precedes Bianchi, EOM, integration-by-parts, trace, and Cayley–Hamilton reduction. It does not show that an $H(4)$ basis cannot exist. It shows that the next construction must first replace the raw carrier by an unpadded/dependent syntax or quotient padding explicitly.

The finite gaps are real but Clay-irrelevant. The $\mathbb{Z}_2$ and $\mathbb{Z}_3$ spectral gaps are honest theorems, but they are abelian, single-link, strong-coupling, and finite. They exemplify the finite OS-reconstruction mechanism; they are not lattice approximations to a continuum $\mathrm{SU}(N)$ theory, and the accompanying scaling diagnostics show that the physical gap can close under a naive continuum schedule. They establish the machinery, not the target.

The continuum mass gap is correctly open. The continuum endpoint is gated `openGoal` with no claimed support, and the constructive-QFT prerequisites are all marked unrepresented. The formalization makes no claim, in either direction, about the existence of a continuum Yang–Mills mass gap.

4 What remains, exactly

Machine-checked since the first version of this report

A second formalization round closed several items that the first version listed as open, and sharpened others into theorems:

- **The two-source bootstrap threshold.** Linear contraction ($\Lambda < 1$) is *not* sufficient for the manuscript's bootstrap; the scalar induction closes iff $a = C\,b^{3-d_{\max}} \leq 1/3$ (`scalar_bootstrap_two_sources`). At $C = 2224$, $b = 2$: depth 14 fails even linear contraction, depth 15 contracts linearly but the bootstrap *fails* (`not_twoSourceBootstrapSlack_2224_two_fifteen`), depth 16 closes. The true reason for $d_{\max} = 16$ is the $1/3$ threshold, not parity of the depth — and the threshold is *dynamical*: an explicit nonlinear toy orbit realizes the recursion with equality and exits the ε^*-ball in one step at depth 15 (`toyRG_ball_escape_two_fifteen`, `benToyRGBootstrapRealization`).
- **The extraction realization and transfer obstruction.** The manuscript's literal displayed sum is ≥ 3 for every $n \geq 1$ and never ≤ 2 (`three_le_literalExtractionOperatorSum`, `not_literalExtractionOperatorSum_le_two`). On an explicit finite Wilson-block polynomial algebra, coefficient-majorant truncation is norm one and gauge invariant (`WilsonBlockMajorant`). A matrix column-sum theorem transfers through explicit basis and dual jets; v14 does not provide these for $P_{\text{ext}} \leq 16$, and a skew-projection counterexample proves that no bound follows from the range alone (`ExtendedExtractionTransfer`). The source-certificate audit further proves that F.4 and O.9 select different EOM relation policies and records the absent lattice representatives, quotient basis, dimension-16 jets, biorthogonality, and conditioning rows (`Dimension16WilsonExtractionCertificate`).
- **The support-growth dichotomy.** The two collar-growth statements the manuscript makes are inequivalent, and both branches are now tight theorems: a coarsen-then-thicken recurrence

$r_{j+1} \le r_j/b + A$ yields the uniform bound $r_0/b^J + Ab/(b-1)$, while a thicken-only recurrence $r_{j+1} \le r_j + A$ yields exactly $r_0 + AJ$ (`support_growth_dichotomy`).

- **Gap $\Leftrightarrow$ clustering, finite class.** Reflection positivity, self-adjointness, and a q-contraction on the vacuum complement give exponential decay of truncated two-point functions at rate $-\log q$ (`abs_truncatedTwoPoint_le_exp`); conversely, RP plus per-vector moment decay $\langle v, T^n v\rangle \le C_v q^n$ on a dense subset gives the gap (`eigenvalue_abs_le_of_moment_decay`) — note the *per-vector* constants: only the rate must be uniform, which weakens what the manuscript's Assumptions K.7/M.5 need to supply. A regression records that an additive t-independent tail defeats the conclusion (`moment_decay_additive_defeat`), so large-field remainders must decay, not merely be small.

- **A non-abelian instance.** The quaternion group $Q_8 \subset SU(2)$ single-link model: machine-checked non-commutativity ($ij = k \neq -k = ji$), reflection positivity, spectral gap carried by the four-dimensional spin isotypic component, and clustering; together with the $\mathbb{Z}_2$ instance this discharges the lattice-side inputs (`massGap`, `reflectionPositive`, `clustering`) of the finite continuum-gate conditional (`z2_continuumMassGap_of_reconstructionMachine`).

Machine-checked in the third round

A third round moved from schematic carriers to the exact SU(2), $b = 2$ Wilson block (§3) and decided the extraction stage on it:

- **Hard truncation refuted intrinsically.** The Cayley–Hamilton relation $a^3b - 4a^4 + a^5 = 0$, with monomial dimensions $(16, 16, 20)$ and the explicit -32 evaluation, proves representative-wise canonical truncation is not well-defined on the evaluated trace-word quotient (`SU2CayleyHamiltonTruncationObstruction`).

- **Exact norm-below-two extractors constructed.** The 21-coordinate associated-graded jet chart has extraction constant 131071/65536 and the one-plaquette trace quotient has 31/16, both certified with explicit bases, identity jet matrices, and biorthogonal duals; the conditional ledger value $90832203/40960 \approx 2217.6$ closes the two-source bootstrap at depth 16 and fails at depth 15 (`SU2AssociatedGradedExtraction`, `SU2IndependentTraceBasis`).

- **A certified dimension separation.** The full graded chart has dimension 12,875,774,670 and admits no linear equivalence onto any space of dimension $\le$ 1851, so v14's schematic operator count requires an invariant quotient the manuscript does not construct (`SU2AssociatedGradedExtractionDecision`).

- **The (F, D) transfer socket is uninhabited as written.** The dimension-18/radial-degree-6 descendant and the integration-by-parts/block witness $\partial^2\mathrm{Tr}(F^2)$ jointly refute the faithful transfer sockets for both relation policies and all RG-compatible extensions (`v14_fullFD_quotientTransfer_noGo`).

Machine-checked in the fourth round

A fourth round constructed the derivative-aware boundary-cochain scaffold and then decided whether its dimension-16 coordinate socket could be filled by the manuscript's Lorentz-only operator census:

- **The repaired filtration and boundary bookkeeping are mechanical.** The canonical-dimension filtration, boundary-aware cochain coordinates, coefficient projection, and conditional scalar bootstrap are machine-checked. Their analytic Wilson-RG factors remain explicit inputs.

- **The actual-lattice O.9 census is refuted at dimension 8.** The exact SU(2) Wilson observable, Cartan chart, fourth-jet calculation, and rational rotation witness prove that the hypercubic quartic jet is not Lorentz invariant (`SU2LatticeFDCensusNoGo`).

- **Both relation branches and six same-target variants are closed.** Neither the off-shell O.9.2 policy nor the on-shell F.4.3 policy admits a complete Lorentz-only certificate. EOM, Bianchi/IBP, cyclic/multitrace, SU(2) Cayley–Hamilton, and postponed-IBP variants share the same uninhabited spanning core.

- **The live fork is explicit.** One must either enumerate an $H(4)$-complete finite-spacing basis and rederive its duals and constants, or prove a Symanzik/continuum projection with quantitative decay of the Lorentz-breaking artifact coordinates. No numerical factor transfers automatically from the refuted census.

Machine-checked in the fifth round

The first branch of the live fork was pursued directly:

- **The exact hypercubic group and finite raw syntax are mechanical.** Signed coordinate permutations form a concrete group of order 384. The raw surface records field and derivative indices, trace cycles, derivative ownership/order, canonical dimension through 16, scalar/pseudoscalar characters, and separate relation-family labels. A label still supplies no relation row by itself (`V14HypercubicFDCensus`).

- **The unpadded dimension-8 quartic census is complete.** The 126 Cartan quartic monomials have checked signed-orbit certificates. Four surviving orbit functions form a basis, representative evaluation supplies exact duals, and the conditioning matrix is the identity (`V14HypercubicQuarticCensusCertificate`, `V14HypercubicQuarticBasis`).

- **The actual Wilson and artifact coordinates are certified.** The Wilson fourth jet is exactly coordinate vector $(6, 0, 0, 0)$. The $O(4)$-invariant function subspace is proper, and the pure-fourth polynomial is nonzero in its quotient (`V14HypercubicQuarticWilsonBridge`).

- **The padded dimension-16 extension is refuted.** Two distinct fixed-width encodings of the empty active syntax differ only in inactive padding. Any active-syntax-faithful basis gives them equal rows, which is incompatible with the right-inverse identity required by `ExactCensusCertificate` (`V14HypercubicFDCensusPaddingNoGo`).

- **The corrected next object is exact.** Replace the padded raw carrier by finite unpadded/dependent (F, D) syntax, or form a checked padding quotient on which the $H(4)$ action and all relation rows descend. Then enumerate the full signed orbits and relation quotient through dimension 16. No dimension-8 constant transfers automatically.

The remaining inputs

With the arithmetic and structural skeleton closed, the route now rests on the following inputs — listed with their exact status in v14, since these are three different situations.

1. **Derive the extraction/recombination constant on the actual map.** *Status: not derived.* Appendix F.5's displayed factors equal 33152 at $b = 2$; Appendix O's different factors with $C_{\mathrm{extract}} = 2$ equal 11088/5. The fluctuation big-O constant, weighted cumulant-generation estimate, rooted tree-graph factor, and basis/dual-jet conditioning are not proved on the actual map. Regrouping connected families by union has norm one; the factor 14 is the local rooted branching count, not a total-contact bound. The third round sharpened what a derivation must overcome: representative-wise truncation is refuted intrinsically, exact norm-below-two extractors exist on the corrected charts, and the transfer of any such extractor to the canonical (F, D) space is refuted by the derivative-blindness and integration-by-parts/block obstructions. *What settles it:* construct a derivative-aware filtration whose truncation provably agrees with canonical dimension $2 \cdot \#F + \#D$, together with a boundary-aware complex retaining total-derivative descendants per block and imposing integration by parts only at global recombination. The fourth round constructed that mechanical scaffold but proved that its actual-lattice coordinate census cannot use only O.9.3's Lorentz scalars. The fifth round completed the unpadded Cartan quartic $H(4)$ basis and identity dual chart, but proved that the fixed-width dimension-16 raw carrier cannot support a faithful invertible census because inactive padding remains observable. *What settles it now:* construct a finite unpadded/dependent (F, D) syntax or a checked padding quotient, prove that the $H(4)$ action and the explicit relation rows descend, and certify its basis and conditioning through dimension 16; alternatively supply a quantitative continuum/Symanzik projection controlling every discarded Lorentz-breaking coordinate. Then prove the norm-below-two estimate, the weighted Wilson cumulant/tree estimate, and the Wilson-RG intertwining law on the same object. The resulting actual constant can then be checked against $C\, b^{3-d_{\max}} \leq 1/3$.

2. **Which support recurrence the glue satisfies.** *Status: v14 asserts both branches of a now-proven dichotomy in different places. What settles it:* one paragraph identifying the recurrence. If coarsen-then-thicken, the uniform bound is already a theorem; if thicken-only, the $O(J)$ growth breaks uniform tree decay and the step needs a new argument.

3. **The two-marked correlation identity.** *Status: missing.* The glue / back-propagation step requires a *two*-marked transport identity of the shape $\langle A; \tau B\rangle_{\mu_0} = \langle QA; \tau' QB\rangle_{\mu_J}$; v14 states the one-marked pushforward duality, and the two-marked version does not follow from it (conditional expectations do not factor). *What settles it:* state it precisely and prove it, or carry it as an explicit named assumption; once stated, its formalization is immediate.

4. **KP convergence at the stopping scale and Wilson-measure reflection positivity.** *Status: abstract results are classical; the route-specific instantiation is open.* One must construct the actual stopping-scale Wilson activities in the same norm, prove the KP criterion uniformly, and verify reflection and boundary-limit compatibility for the same Wilson family and RG kernels.

5. **The real RG map satisfies the recursion.** *Status: the central claimed content of v14's RG analysis; not independently verified here.* Everything above shows the surrounding machine is consistent and sharp; whether the actual Yang–Mills RG map, with its actual constants,

satisfies the two-source recursion inside the 1/3-threshold is the mathematical core of the route — the one input that is neither bookkeeping, nor a clarification, nor classical.

Items (1), (2), (3), and (5) contain route-specific mathematics. The classical theorem statements in item (4) do not discharge their Wilson instantiation. The actual RG recursion and constructive-QFT promotion remain open, and the Yang–Mills mass gap remains open.

Lean modules. The results reported here are in `Mettapedia.QuantumTheory.YangMills.RGBootstrap` (the contraction predicate and knife-edge arithmetic), `Mettapedia.QuantumTheory.YangMills.RGCrux` (the even lower-extraction route obstruction), `Mettapedia.QuantumTheory.YangMills.FiniteOSReconstruction`, `Mettapedia.QuantumTheory.YangMills.Z2StrongCouplingGap`, and `Mettapedia.QuantumTheory.YangMills.Z3StrongCouplingGap` (the finite OS reconstruction and strong-coupling gaps), and `Mettapedia.QuantumTheory.YangMills.ProofState` (the continuum-open gating); and, from the second round, `RGBootstrapSlack` (two-source bootstrap and depth trichotomy), `ExtractionMajorant` (displayed-sum audit and generic majorant model), `WilsonBlockMajorant` (the concrete finite Wilson-polynomial realization), `ExtendedExtractionTransfer` (the range-only obstruction and explicit basis/dual-jet transfer theorem), `Dimension16WilsonExtractionCertificate` (the F.4/O.9 relation-policy mismatch, checked source transfer table, and obstructed as-written coordinate certificate), `WilsonPolymerRecombination` (norm-one union regrouping, the total-contact counterexample, and the rooted interpretation of 14), `WilsonExtractionRecombinationConstant` (the two ledger calculations, analytic-factor obstructions, and conditional depth arithmetic), `SupportGrowth` (the recurrence dichotomy), `TransferGapClustering` and `Z2TransferClustering` (gap $\Leftrightarrow$ clustering and the discharged finite continuum gate), `ToyRGMap` (the dynamical realization and depth-15 escape), and `Q8StrongCouplingGap / Q8TransferClustering` (the non-abelian instance); and, from the third round, `SU2WilsonBlockPilot` (the exact SU(2) block, tree gauge, co-tree coordinates, and fixed-boundary Haar reference integral), `SU2CayleyHamiltonTruncationObstruction` (the intrinsic non-descent witness), `SU2MajorantRepairDecision` (the sealed truncation-repair decision), `SU2AssociatedGradedExtraction / SU2AssociatedGradedExtractionDecision` (the graded jet basis, its exact extraction constant, and the dimension separation from the schematic count), `SU2IndependentTraceBasis` (the one-plaquette normal form and its extraction constant), and `V14FDQuotientTransferNoGo` (the derivative-blindness and integration-by-parts/block obstructions and the uninhabited transfer sockets); and, from the fourth round, `V14BoundaryCochainComplex`, `V14BoundaryCochainExtraction`, and `V14BoundaryCochainBootstrap` (the derivative-aware boundary-cochain scaffold and its conditional arithmetic), together with `SU2LatticeFDCensusNoGo` (the exact hypercubic quartic witness and the off-shell/on-shell Lorentz-census no-go); and, from the fifth round, `V14HypercubicFDCensus` (the signed $H(4)$ action and finite raw surface), `V14HypercubicQuarticCensusCertificate` and `V14HypercubicQuarticBasis` (the complete unpadded Cartan quartic census, duals, and identity conditioning), `V14HypercubicQuarticWilsonBridge` (the exact Wilson coordinate vector and nonzero $O(4)$-artifact quotient class), and `V14HypercubicFDCensusPaddingNoGo` (the uninhabited faithful padded dimension-16 exact census) — all in the same namespace, all axiom-guarded to {`propext`, `Classical.choice`, `Quot.sound`} with no `sorry`. The primary mathematical source is Ben Goertzel's manuscript *Yang–Mills Mass Gap in 4D: RG Bootstrap, Polymer Expansion, OS Reconstruction* (v14).